How-To Library

Informatica

# Configuring a Hadoop Environment for Test Data Management

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# Abstract

You must install and configure a Hadoop environment if you want to perform Test Data Management (TDM) operations with Hadoop connections.

The article describes how to install a Hadoop environment, configure the Data Integration Service, and configure Hive and Hadoop Distributed File System (HDFS) connections.

# Supported Versions

- Test Data Management 9.7.0
- Test Data Management 9.7.1

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# Overview

You can perform data masking, data domain discovery, and data movement operations on Big Data Edition Hadoop clusters.

You must install a Hadoop environment for TDM. The Informatica Big Data Edition installation is distributed as a Red Hat Package Manager (RPM) installation package. The RPM package and the binary files that you need to run the Big Data Edition installation are compressed into a tar.gz file.

# Configure Hadoop Environment

In TDM, you can use Hive or HDFS connections as sources or targets.

Create Hive or HDFS connections in Test Data Manager to perform data masking, data domain discovery, and data movement operations.

To configure Hadoop environment for TDM operations, perform the following steps:

1. Install RPM on Hadoop.
2. Configure Hadoop cluster properties.
3. Configure Hadoop pushdown properties for the Data Integration Service.
4. Configure Hadoop connections.
5. Configure Hive properties.

## Step 1. Install RPM on Hadoop

You can install the RPM package for Hadoop on a single node or on multiple node clusters.

1. Install Informatica RPM on the Hadoop machine that you want to use as the target.
2. If there are multiple nodes, install the RPM on all the nodes of the cluster.

   The installation path must be same in all the nodes of the cluster. For example, you can install RPM in the `\opt` folder in all the nodes of the cluster.

## Step 2. Configure Hadoop Cluster Properties

Configure Hadoop cluster properties in the `yarn-site.xml` file that the Data Integration Service uses when it runs mappings on a Cloudera CDH cluster or a Hortonworks HDP cluster.

1. Copy the `yarn-site.xml` file from the Hadoop cluster to the following location:

   ```
   <Informatica installation directory>/services/shared/hadoop/<Hadoop_distribution_name>/conf/
   ```

2. Ensure that the following properties are present in the `yarn-site.xml` file that you copied:

   ```
   <property>
   <name>mapreduce.jobhistory.address</name>
   <value><NAMENODE>:10020</value>
   <description>MapReduce JobHistory Server IPC host:port</description>
   </property>
   <property>
   <name>mapreduce.jobhistory.webapp.address</name>
   <value> <NAMENODE>:19888</value>
   <description>MapReduce JobHistory Server Web UI host:port</description>
   </property>
   <property>
   <name>yarn.resourcemanager.scheduler.address</name>
   <value> <NAMENODE>:8030</value>
   <description>CLASSPATH for YARN applications. A comma-separated list of CLASSPATH entries</
   description>
   </property>
   ```

3. Copy the `hive-site.xml` file from the Hadoop cluster to the following location:

   ```
   <Informatica installation directory>/services/shared/hadoop/<Hadoop_distribution_name>/conf/
   ```

4. Ensure that the following properties are updated in the `hive-site.xml` file that you copied:

   ```
   <property>
   <name>hive.metastore.uris</name>
   <value>thrift://<NAMENODE>:9083</value>
   <description>Thrift uri for the remote metastore. Used by metastore client to connect to
   remote metastore.</description>
   </property>
   <property>
   ```

```
<name>mapreduce.jobhistory.webapp.address</name>
<value><NAMENODE>:19888</value>
</property>
<property>
<name>fs.defaultFS</name>
<value>hdfs://<NAMENODE>:8020</value>
</property>
<property>
<name>mapreduce.jobhistory.address</name>
<value><NAMENOODE>:10020</value>
</property>
```

5. Verify that the ODBC entry files, TNS entry files, and DB2 installation entries are specified in the following location: `<Informatica installation directory>/services/shared/hadoop/<Hadoop_distribution_name>/infaConf/hadoopEnv.properties`

   The following example shows the environment variables that you can edit:

   ```
   infapdo.env.entry.ld_library_path=LD_LIBRARY_PATH=$HADOOP_NODE_INFA_HOME/services/shared/bin:$HADOOP_NODE_INFA_HOME/DataTransformation/bin:/opt/teradata/client/14.10/tbuild/lib64:/opt/teradata/client/14.10/odbc_64/lib:/databases/oracle_11.2.0/lib:/databases/db2v9.5_64BIT/lib64:$HADOOP_NODE_HADOOP_DIST/lib/native:$HADOOP_NODE_INFA_HOME/ODBC7.1/lib:$HADOOP_NODE_HADOOP_DIST/lib/native/Linux-amd64-64:$LD_LIBRARY_PATH
   infapdo.env.entry.path=PATH=$HADOOP_NODE_HADOOP_DIST/scripts:$HADOOP_NODE_INFA_HOME/services/shared/bin:/databases/oracle_11.2.0/bin:/databases/db2v9.5_64BIT/bin:$HADOOP_NODE_INFA_HOME/ODBC7.1/bin:$PATH
   infapdo.env.entry.oracle_home=ORACLE_HOME=/databases/oracle_11.2.0/
   infapdo.env.entry.tns_admin=TNS_ADMIN=/opt/ora_tns
   infapdo.env.entry.db2_home=DB2_HOME=/databases/db2v9.5_64BIT
   infapdo.env.entry.db2instance=DB2INSTANCE=db95inst
   infapdo.env.entry.db2codepage=DB2CODEPAGE="1208"
   infapdo.env.entry.odbchome=ODBCHOME=$HADOOP_NODE_INFA_HOME/ODBC7.1
   infapdo.env.entry.odbcini=ODBCINI=/opt/ODBCINI/odbc.ini
   ```

6. When you install on multiple nodes of a cluster, copy the `hdfs-site.xml`, `core-site.xml`, and `mapred-site.xml` files from the `/usr/lib/hadoop/conf` cluster to the `<Domain_installation>/services/shared/hadoop/[hadoop_Distribution]/conf` cluster.

## Step 3. Configure Hadoop Pushdown Properties for the Data Integration Service

Configure Hadoop pushdown properties for the Data Integration Service to run mappings in a Hive environment.

1. Log in to the Administrator tool.
2. In the **Manage | Services and Nodes** view, select the Data Integration Service in the domain from the Navigator pane.
3. Click the **Processes** tab on the right pane.
4. In the **Execution Options** section, configure the following properties:

**Informatica Home Directory on Hadoop**

The Big Data Edition home directory on every data node created by the Hadoop RPM install. Type `/<HadoopInstallationDirectory>/Informatica`

**Hadoop Distribution Directory**

The directory that contains a collection of Hive and Hadoop JARS on the cluster from the RPM install locations. The directory contains the minimum set of JARS required to process Informatica mappings in a Hadoop environment. Type `/<HadoopInstallationDirectory>/Informatica/services/shared/hadoop/<Hadoop_distribution_name>`

**Data Integration Service Hadoop Distribution Directory**

The Hadoop distribution directory on the Data Integration Service node. The contents of the Data Integration Service Hadoop distribution directory must be identical to the Hadoop distribution directory on the data nodes. The Hadoop Distribution at the end specifies which jars have to be used while running the mappings in Hadoop and Data Integration Service modes. The Hadoop RPM installs the Hadoop distribution directories in the following path: `<Informatica installation directory>/services/shared/hadoop/<Hadoop_distribution_name>`

5. Restart the Data Integration Service.

   **Note:** When you create the Data Integration Service, the Mapping Service Module is enabled.

# Step 4. Configure Hadoop Connections

After you install RPM on a Hadoop machine and configure the Data Integration Service, you must create Hadoop connections in Test Data Manager.

You can create Hive or HDFS connections to perform TDM operations.

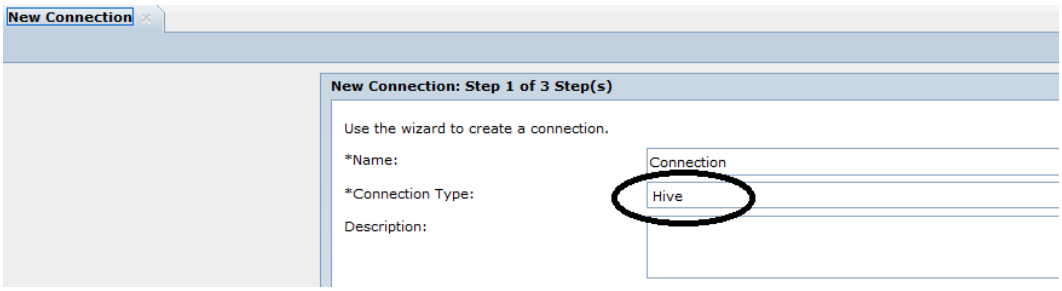
## *Creating a Hive Connection*

In Test Data Manager, create a Hive connection and use the connection as a source or a target when you perform TDM operations.

1. Log in to Test Data Manager.
2. Click **Administrator** > **Connections**.
3. Click **Actions** > **New Connection**.

   The **New Connection** wizard appears with the connection properties.
4. Select the **Hive** connection type.
5. Enter the connection name, description, and owner information.

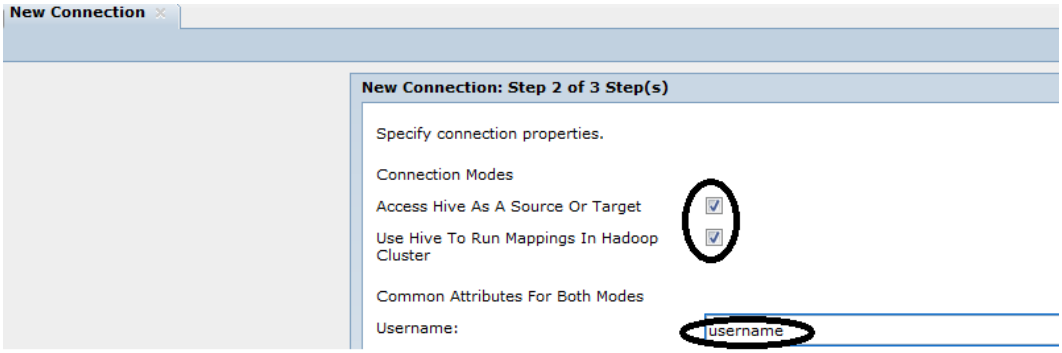
   The following image shows the **New Connection** wizard parameters:




6. Click **Next**.

7. To use Hive as a source or a target, select **Access Hive as a source or target**.
8. To use the connection to run mappings in the Hadoop cluster, select **Use Hive to run mappings in Hadoop cluster**.
9. To access the Hive database, enter the user name.

   The following image shows the connection modes and attributes that you can configure for the Hive connection:

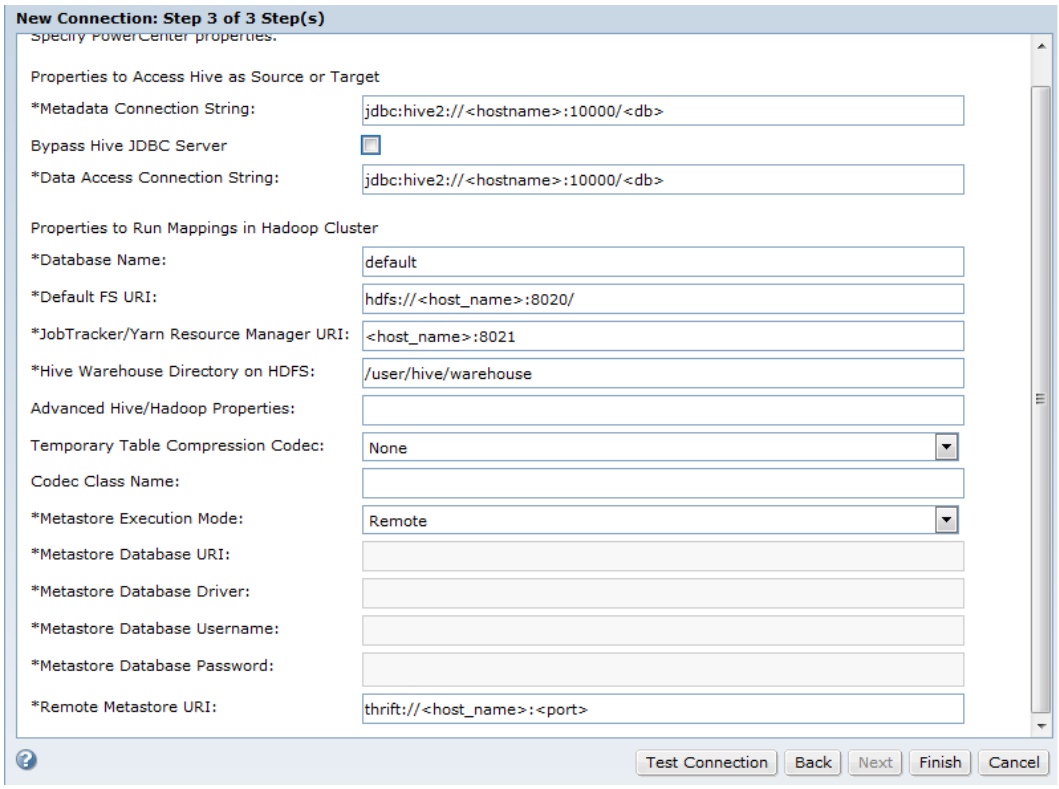



10. Click **Next**.
11. To access the metadata from the Hadoop cluster, enter the metadata connection string in the following format: `jdbc:hive2://<NodeName>:10000/default`.

    For example: `jdbc:hive2://ivlhdp35:10000/default`
    You can also create a schema and provide the schema name instead of the default schema.
12. To access data from the Hadoop cluster, enter the data access connection string in the following format: `jdbc:hive2://<NodeName>:10000/default`

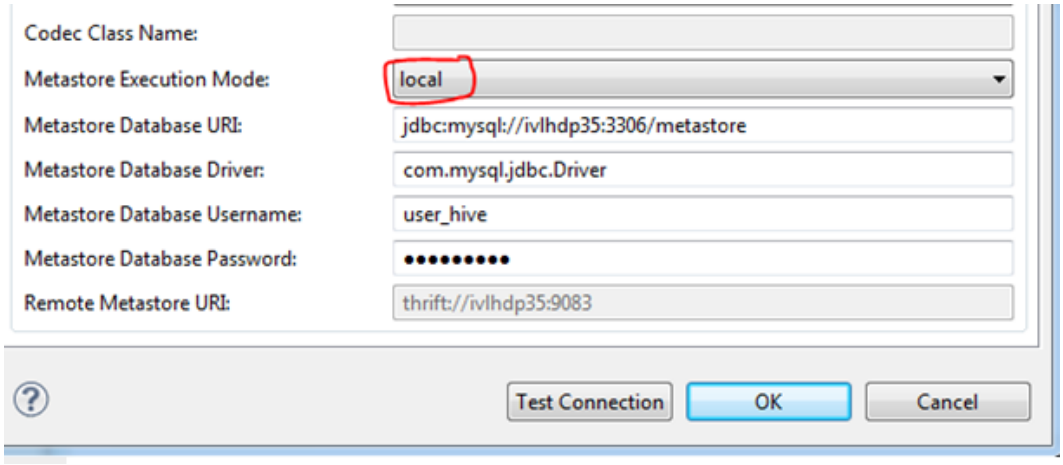
    For example: `jdbc:hive2://ivlhdp35:10000/default`
    You can also create a schema and provide the schema name instead of the default schema.
13. To run mappings in the Hadoop cluster, enter the following parameters:
    - Database Name. Enter the name default for tables that do not have a specified database name.
    - Default FS URI. Enter the URI to access the default HDFS in the following format: `hdfs://<NodeName>:8020/`
      For example: `hdfs://ivlhdp35:8020`
    - JobTracker/Yarn Resource Manager URI. Enter the specific node in the Hadoop cluster in the following format: `<NodeName>:<Port>`.
      For Cloudera, the port is 8032, and for Hortonworks, the port is 8050.
    - Hive Warehouse Directory on HDFS. Enter the HDFS file path of the default database.
      For example, the following file path specifies a local warehouse: `/user/hive/warehouse`
14. To access a Hive metastore, select Local or Remote.
    - Remote. Connects to the thrift server which in turn interacts with the Hive server.
    - Local. Uses a JDBC connection string to connect directly to the MySQL database. Default is Local.
15. To connect to a remote metastore, select **Remote**. If you select Remote, specify only the Remote Metastore URI with the thrift server details in the following format: `thrift://<NodeName>:9083`
    For example: `ivlhdp35:9083`

The following image shows the Hive properties that you can configure:




16. If you select Local mode, specify the Metastore Database URI, driver, user name, and password.

    The following image shows the local metastore execution mode properties that you can configure:




17. To test the connection, click **Test Connection**.
18. To save the connection, click **OK**.

    The connection is visible in the **Administrator | Connections** view.

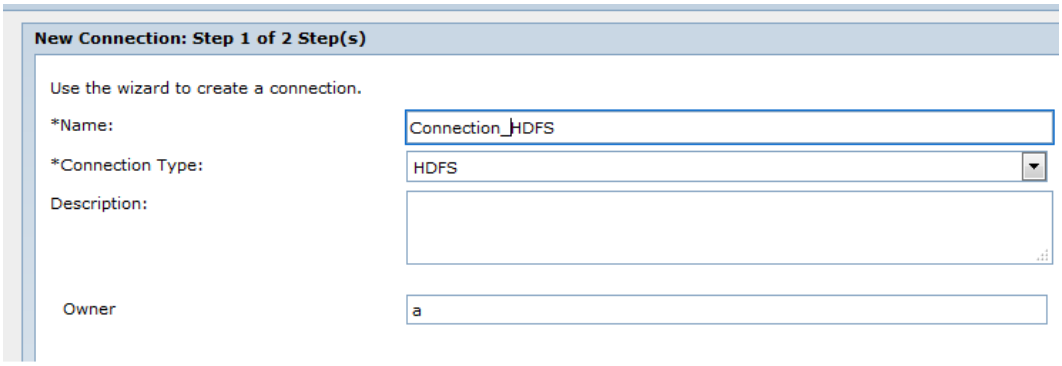
## Creating an HDFS Connection

In Test Data Manager, create an HDFS connection and use the connection as a source or a target when you perform TDM operations.

1. Log in to Test Data Manager.
2. Click **Administrator** > **Connections**.
3. Click **Actions** > **New Connection**.

   The New Connection wizard appears with the connection properties.
4. Select the **HDFS** connection type.
5. Enter the connection name, description, and owner information.

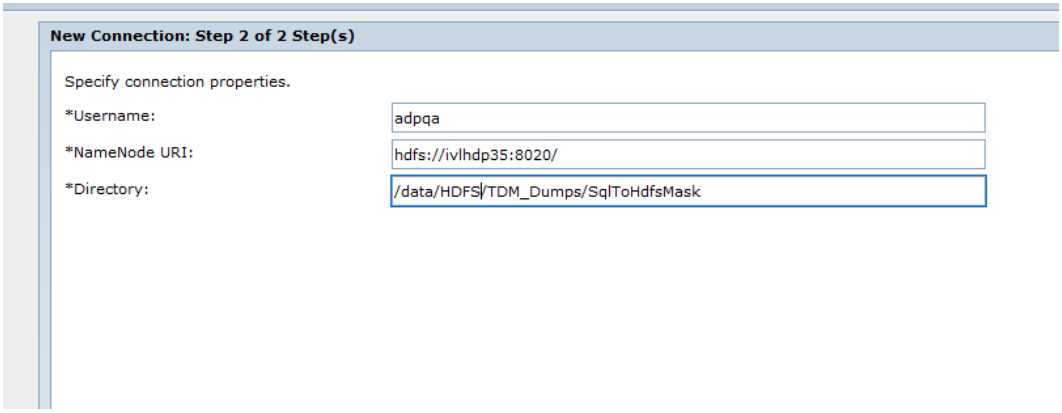
   The following image shows the New Connection wizard with the HDFS connection parameters:




6. Click **Next**.
7. To access HDFS, enter the user name.
8. To access the HDFS URI, enter the NameNode URI in the following format: `hdfs://<namenode>:8020`

   HDFS runs on port 8020.
9. Enter the directory for the Hadoop instance on which you want to perform TDM operations.

   The following image shows the connection properties that you can configure for the HDFS connection:




10. To test the connection, click **Test Connection**.
11. To save the connection, click **OK**.

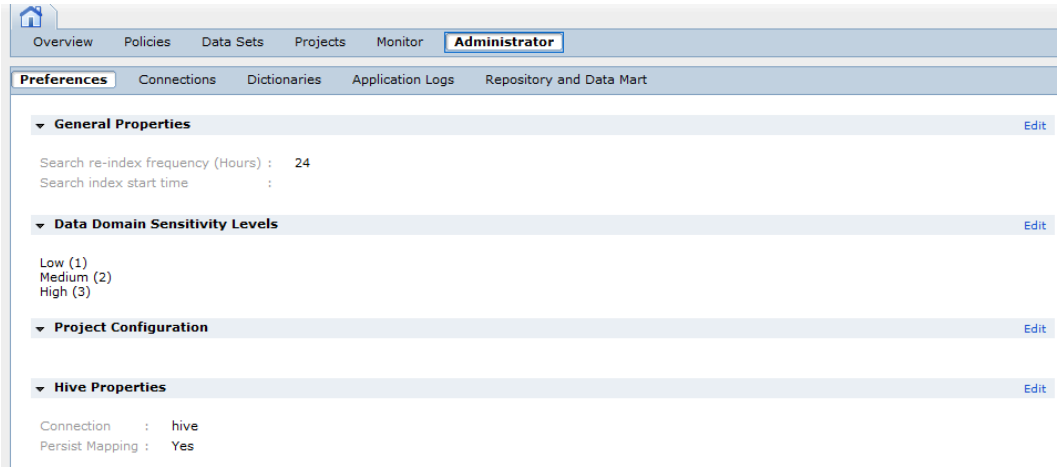
    The connection is visible in the **Administrator | Connections** view.

## Step 5. Configure Hive Properties

To run the mappings from TDM, you must configure the Hive pushdown connection.

1. Click **Adminstrator** > **Preferences**.
2. In the **Hive Properties** section, click **Edit**.
3. Select the **Hive** connection.
4. To view the mappings in the Data Integration Service of the Administrator tool, enable **Persist Mapping**.

   The following image shows the Hive properties that you can configure:




   You can now perform data masking, data movement, and data domain discovery operations on Hadoop connections.

## Author

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